

Lecture 12

Reliability and Variations in VLSI

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Latest Apple Silicon (largest M5 processor) is rumoured to have over 140 billion transistors. For an Apple computer to work reliably, all these transistors must all be working. Anthropologists estimate that in the history of the human race, there may be a sum total of 117 billion people ever lived. Most of them are already dead. This put into perspective how enormous the tasks to design and manufacture a modern VLSI chip.

Many obstacles are in the way:

- Transistor feature sizes are so small that they are below the wavelength of light making photolithography really challenging.
- Number of dopant atoms are now so few that variation across different transistors become significant.
- To avoid breakdown of insulation, the operating voltage of VLSI chips keep coming down reducing noise margin.
- Variations due to temperature changes become significant.
- Devices and interconnects degrade over time particularly under more stressful operating conditions.

This lecture examines the factors that affect the robustness of our designs and how we could minimize their impact on the functionality and longevity of VLSI chips.

Reliability in VLSI

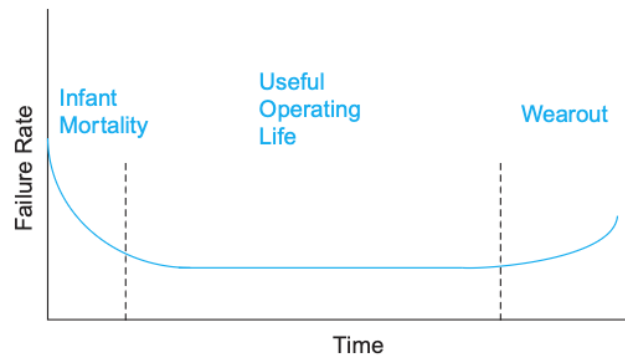
◆ Hard Errors

- Oxide wearout
- Interconnect wearout
- Overvoltage failure
- Latchup

◆ Soft Errors

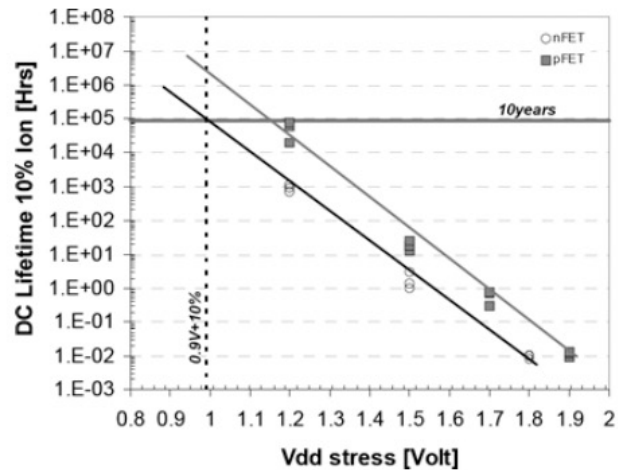
◆ Characterizing reliability

- Mean time between failures (MTBF)
 - # of devices x hours of operation / number of failures
- Failures in time (FIT)
 - # of failures / thousand hours / million devices



Accelerated Lifetime Testing

- ◆ Expected reliability typically exceeds 10 years
- ◆ But products come to market in 1-2 years
- ◆ Accelerated lifetime testing required to predict adequate long-term reliability



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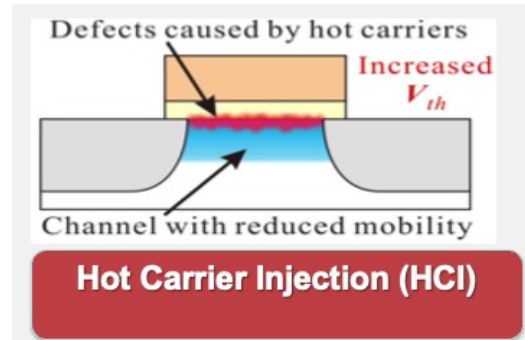
Expected lifetimes of chips is usually more than 10 years. However, it is impossible to test for this long. One technique is to accelerate failure by elevating the supply voltage and operating temperature.

Systems are subjected to accelerated life testing during burn-in conditions to simulate the aging process and evaluate the time to wearout. The results are extrapolated to normal operating conditions.

The plot above shows the measured lifetime of gate oxides in an IBM 32 nm process at elevated voltages [Arnaud08]. The extrapolated results show a lifetime exceeding 10 years at 10% above the nominal 0.9V V_{DD} .

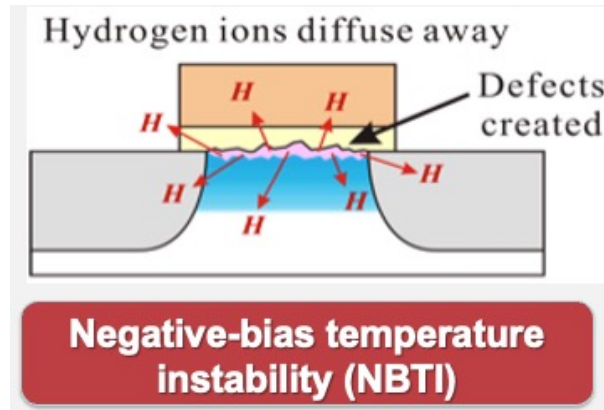
Hot Carriers

- ◆ Electric fields across channel impart high energies to some carriers
 - These “hot” carriers may be blasted into the gate oxide where they become trapped
 - Accumulation of charge in oxide causes shift in V_t over time
 - Eventually V_t shifts too far for devices to operate correctly
- ◆ Choose V_{DD} to achieve reasonable product lifetime
 - Worst problems for inverters and NORs with slow input risetime and long propagation delays



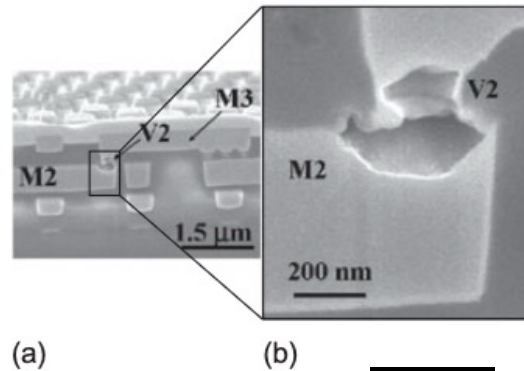
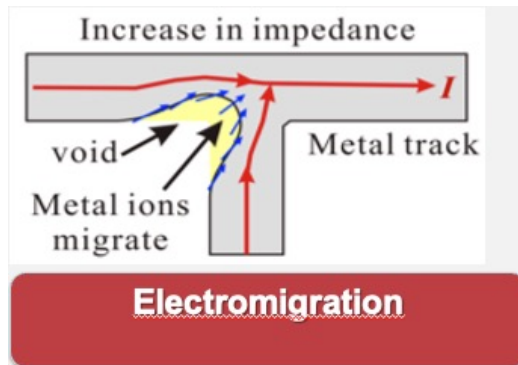
Negative-bias Temperature Instability (NBTI)

- ❑ Electric field applied across oxide forms dangling bonds called traps at Si-SiO₂ interface
- ❑ Accumulation of traps causes V_t shift
- ❑ Most pronounced for pMOS transistors with strong negative bias ($V_g = 0$, $V_s = V_{DD}$) at high temperature



Electromigration

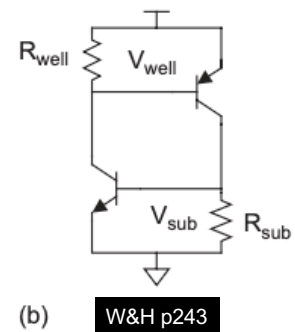
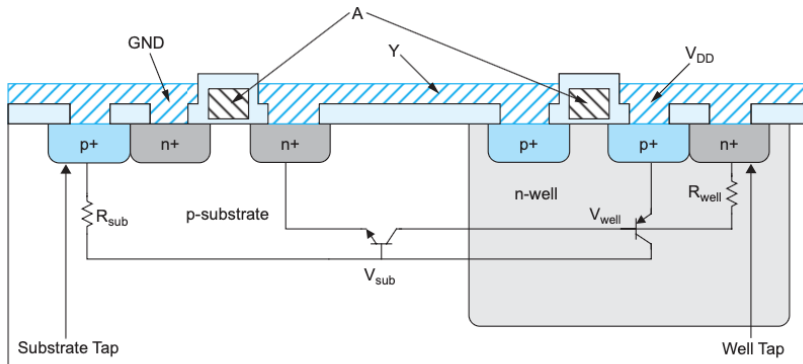
- ◆ “Electron wind” causes movement of metal atoms along wires
- ◆ Excessive electromigration leads to open circuits
- ◆ Most significant for unidirectional (DC) current
 - Depends on current density J_{dc} (current / area)
 - Exponential dependence on temperature
- ◆ Typical limits: $J_{dc} < 1 - 2 \text{ mA} / \mu\text{m}^2$



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Latchup

- ◆ Latchup: positive feedback leading to V_{DD} – GND short
 - Major problem for 1970's CMOS processes before it was well understood
- ◆ Avoid by minimizing resistance of body to GND / V_{DD}
 - Use plenty of substrate and well taps



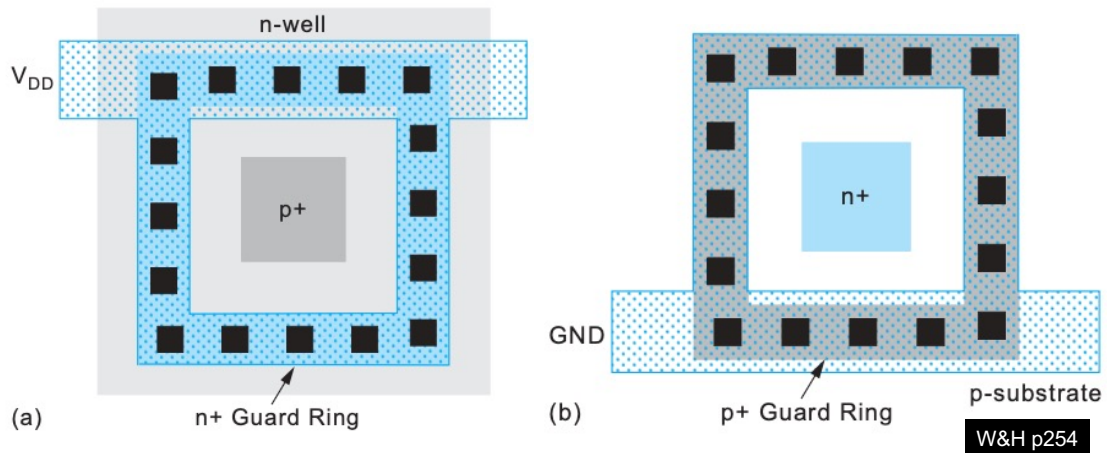
(a)

(b)

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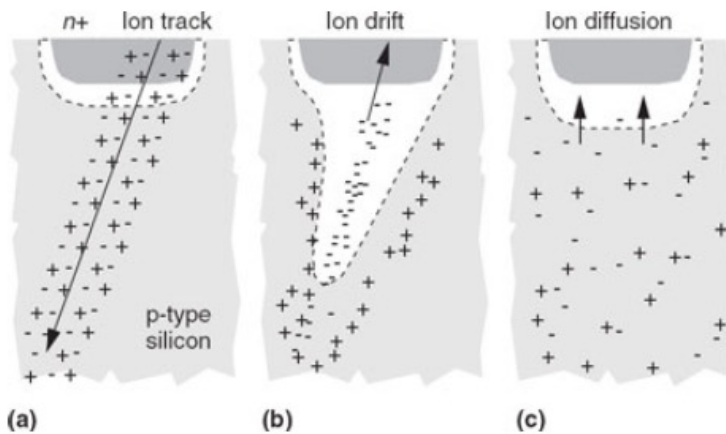
Latchup prevention with Guard Rings

- ◆ Latchup risk greatest when diffusion-to-substrate diodes could become forward-biased
- ◆ Surround sensitive region with guard ring to collect injected charge



Soft Errors

- ◆ In 1970's, DRAMs were observed to randomly flip bits
 - Ultimately linked to alpha particles and cosmic ray neutrons
- ◆ Collisions with atoms create electron-hole pairs in substrate
 - These carriers are collected on p-n junctions, disturbing the voltage



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PVT Variability in VLSI chips

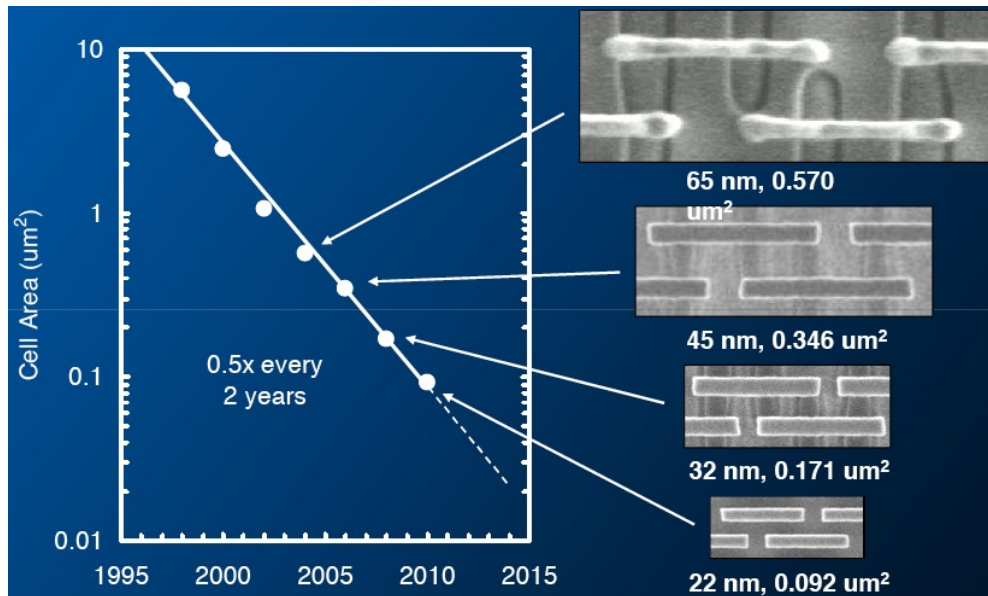
- ◆ There are three main causes :
 1. **Process** variation affecting (P):
 - Threshold voltages
 - Channel lengths
 - Wire dimensions
 2. Environment:
 - Operating **voltage** (V)
 - Operating **temperature** (T)
 3. Aging or wearout
 - Degradation
 - Electron migration

W&H p241

Variation in VLSI could be due to the manufacturing process (P), variation in power supply voltage (V) and the temperature that the chip operating under (T), together, they are known as PVT variations.

In addition, like all things, VLSI chips do age and wear out over time. Device characteristic changes over time and interconnection wires do degrade with a phenomenon known as electron migration, resulting in wires getting thinner!

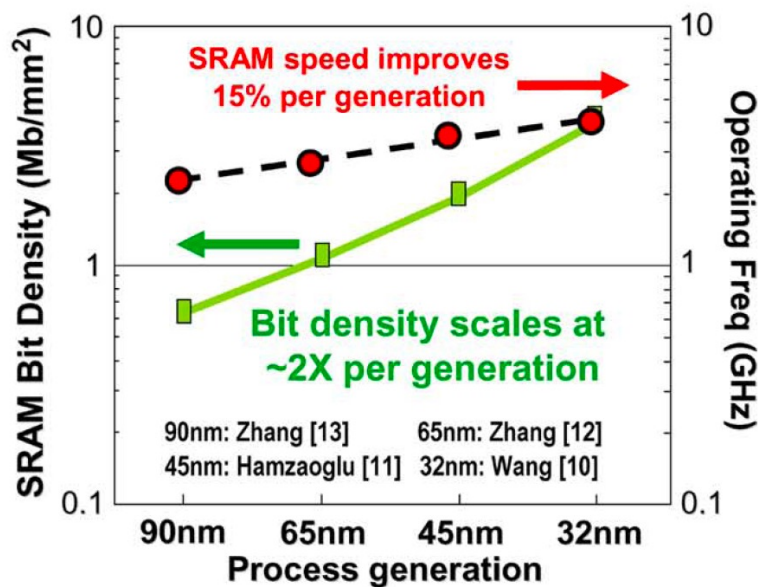
CMOS digital circuits are getting smaller



Source: Intel presentation by Singh, ISPD, 2011

Technology scaling is relentless. This slide from Intel shows how SRAM cells have continually been downsized from 65nm to 22nm, shrinking the cell area by half every technology node every two years. In spite of many previous speculations that such technology scaling will eventually hit a road block, the semiconductor industrial continue to provide advancement along the predict curve for decades. It is likely that progress to smaller geometry will continue for a few more technology nodes in the future.

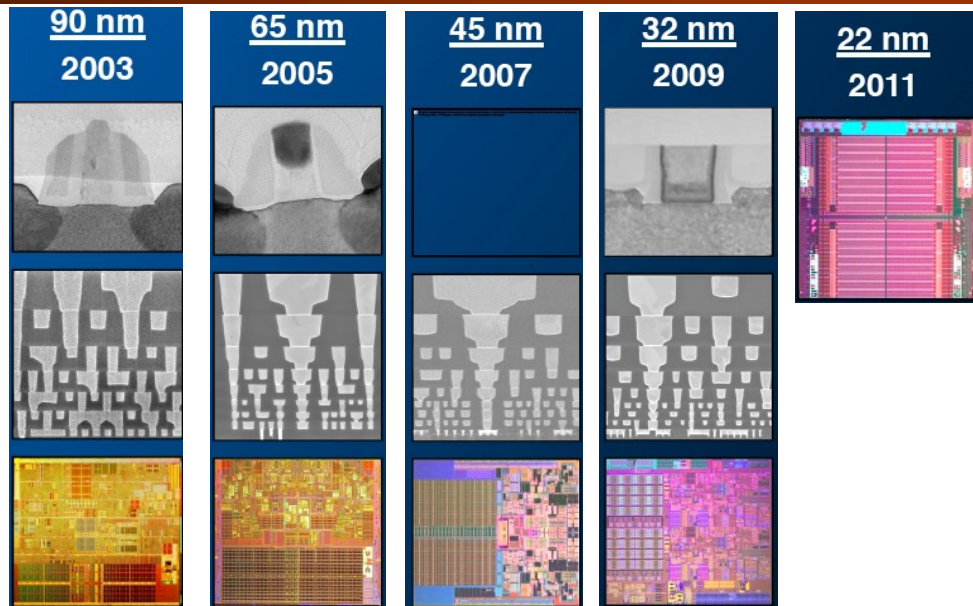
Digital CMOS is also getting faster and denser



Source: Wang et al., "A 4.0GHz 291Mb Voltage-Sacable SRAM Design....", IEEE JSSC (45)1 2010.

As the density of circuit increases, the speed of operation also increases, but at a slower pace. For example while SRAM density increases by a factor two every generation, it only gets a 15% increase in operating frequency. It has now been accepted that future shrinking of feature sizes may not increase by much the effect clock rate as shown in the plot here. 3GHz may be the limit for even handcrafted full-custom design such as those found in Intel microprocessors. Instead, higher performance will come from the many-core approach based on higher degree of parallelism.

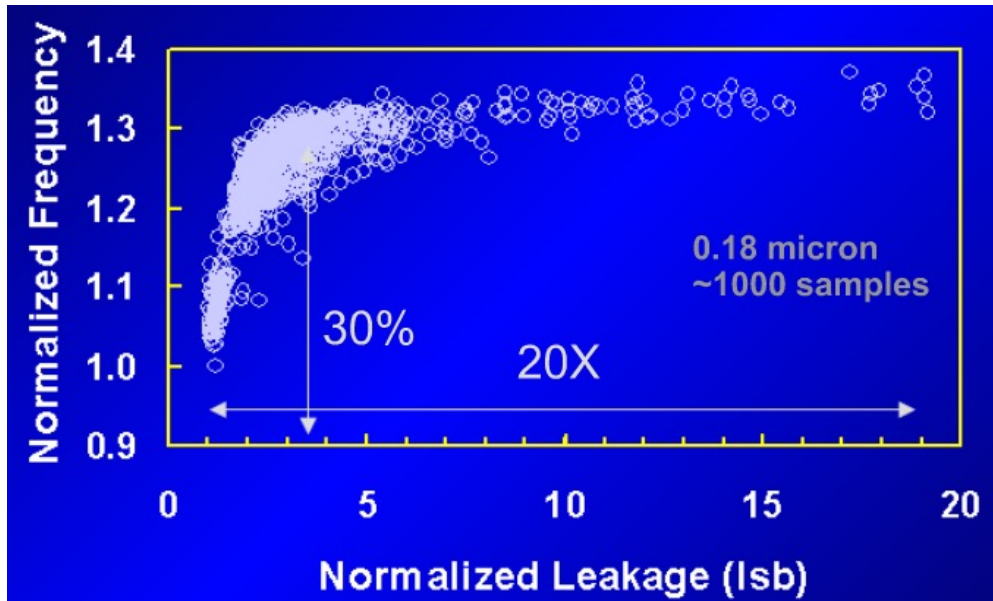
... and it is happening every TWO years!



Source: Intel presentation by Singh, ISPD, 2011

This slides shows the impact of technology scaling to processors instead of SRAMs. As can be seen here, 32nm chips have high number of interconnect layers where the transistor features are more uniform and regular. Corners are sharper and lines are more straight. In a later slide, we will examine the techniques used to achieve such technology scaling.

But process scaling causes large variations

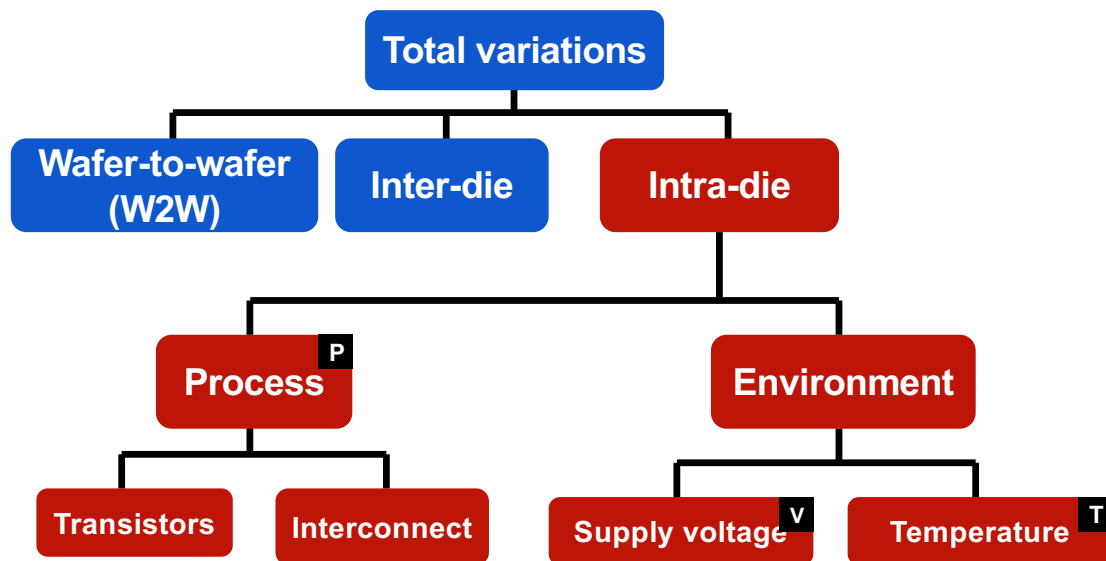


Source: Borkar et al. "Parameter Variations and Impact on Circuits and Microarchitecture", DAC 2003

However, scaling down feature sizes has an undesirable side effect. Here is a plot of operating frequency (normalised to the slowest part) vs normalised leakage current for 1000 samples of chips in 180nm technology. Already it can be seen that with this relatively “tame” technology, the variation in frequency could be as much as 30%, and the leakage current could vary by 20 times! It is expected that such variations will only increase with smaller technology nodes.

This work was first published (by Intel) at DAC in 2003 and since then the problem of variability captured much attention in both industry and academia. This is why we want to spend two hours discussing the problem of variability!

The top-level view on variations



Variations in integrated circuits can be divided in the following types. There are variations from wafer to wafer. Such variation is of no interest to designers and minimizing such variations is the job for the process engineers.

Within a wafer, each die may be different. Such variations are normally handled through speed or leakage binning – part of the manufacturing steps where dies are sorted into performance bins, resulting in different speed or power grades. This is also a topic not really of interest to a designer.

The third category is variation within die (intra-die). This is the type of variation that a designer much worry about. This variations can further be group into two sources: 1) due to the processing of the chip, 2) due to the environment that the chip is operating under. This variability is often known as PVT variations: P = process, V=voltage, T= temperature.

Variations – P, V, T

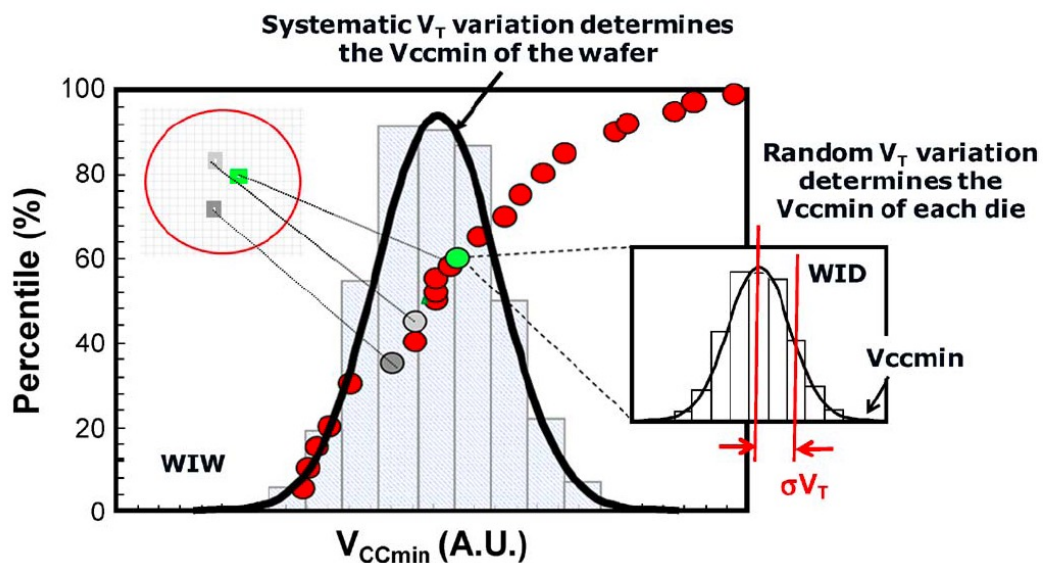
- ◆ **Process** variations – due to the manufacturing process, particularly with technology scaling
- ◆ **Voltage** variations – due to changes in the supply voltage, the variations in IR-drops on board and on-chip, current fluctuations with changing of loads etc.
- ◆ **Temperature** variations – due to changes in environment temperature and/or on-chip temperature, e.g. local hotspots

Process variations are caused by many factors during manufacturing. These have become real challenges to process engineers because the sources of variability have grown significantly as technology scaling advances. More and more processing steps have been introduced, each adding to more variations. Variability is one of the price that has to be paid by scaling down feature sizes.

Voltage variations is the consequence of two factors: scaling down of supply voltages and reducing interconnect width, thus increasing the IR-drop on chip and on-board. Denser transistors also means that more are being packed into a small area, thus increasing the current density on interconnects.

Finally, temperature variations is also becoming more significant. As temperature rises, electrons move slower and therefore delay increases. Higher circuit density also increases both the power dissipation and create uneven temperature distribution on the chip (such as local hot spots).

Inter-die vs Intra-die variations



Source: Wang et al., "A 4.0GHz 291Mb Voltage-Sacable SRAM Design....", IEEE JSSC (45)1 2010.

Variations in integrated circuit can also be divided into two types: systematic variations and random variations.

Systematic variations used to be the dominant type in integrated circuits. It is caused by the imperfection of the fab process such as non-uniform distribution of dopants over the wafer. In the plot above, it shows how the threshold voltage V_{th} varies over the entire wafer as manifested in the minimum operating V_{CC} (i.e. how low can the supply voltage be with the SRAM still working). For example, the three dies on the wafer are picked out: dark grey, light grey and green. They are located in different part of the wafer. The fact that various dies have different $V_{CC}(min)$ is due to systematic variation of V_{th} across the wafer.

If we now zoom into the green die, within this die, you will find that transistors have different V_{th} values, which is also approximately Gaussian distributed. This is the random V_{th} variations. Here the worst-case transistor will determine the $V_{CC}(min)$ of this particular die.

Sources of Process Variations – Old and New

- ◆ Historical Sources:
 1. Lithographical variations
 2. Line roughness
 3. Gate dielectric variations (e.g. oxide thickness, defects and traps)
- ◆ Emerging Sources (those becoming more important due to scaling):
 4. Random dopant fluctuation (RDF)
 5. Implants and annealing variations
 6. Strain variations (due to new process steps such as high-stress capping layers)
 7. Gate material granularity

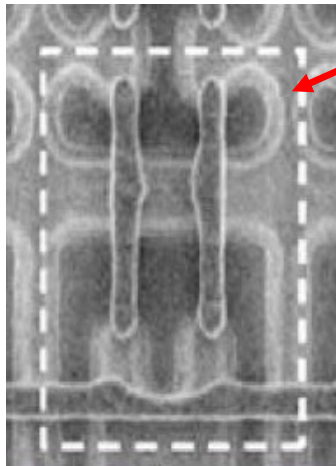
Sources of process variations can be divided into the “historical” sources. These have been around for a long time. Technology scaling for each node requires new techniques to overcome each of these limitations. They are:

1. Lithographical variations – these are variations due to the imperfection in the lithographic process such as the mask production, the photo-resist variations etc.
2. Line roughness – features in most digital ICs are rectilinear (i.e. in straight lines). Unfortunately fabricated lines have unevenness that results in variations.
3. Gate structures also have imperfections. In slide 5, it is clear that earlier technologies have irregular gate structure, which results in variation in parameters such as V_{th} and I_{sd} .

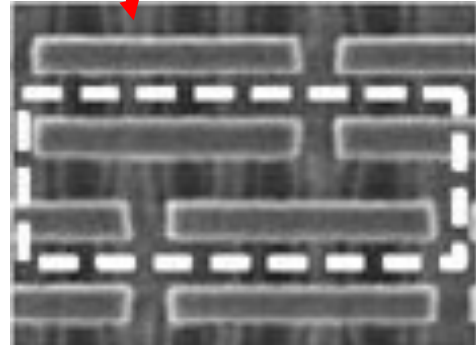
There are also new sources due to the shrinking of feature sizes. These are:

4. Random dopant fluctuations – we will examine this in details later.
5. Implant and annealing steps also have variations.
6. In an effect to improve speed of switching, strain silicon are used, which add more variations.
7. New gate materials such as high-k dielectric (to mitigate high leakage current) add more variations.

Variations Sources – Line and feature roughness



- ◆ 90 nm SRAM cell (left) with significant roughness and unevenness in lines features, corners and shapes
- ◆ Contrast this with a 32nm SRAM cell showing clear improvements in roughness, thus reducing variations

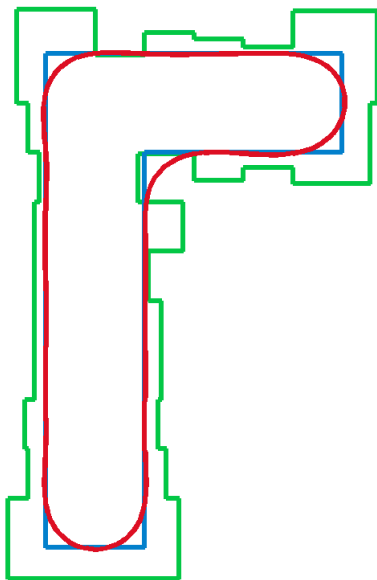


Source: Kuhn et al. "Process Technology Variations", T-ED (58), 8, 2011

Here is a clear example of line roughness type of variations. The left SEM photo shows the fabricated SRAM cell at 90nm. Note the roughness of the lines and the roundness of the corners. The irregularity of these features causes variations. If these were shrunk with the same degree of roughness, the resulting transistor would either not work at all (e.g. with a break in the feature) or will have such large variability that render the device no usable.

The right photo shows a 32nm SRAM cell. In an effort to reduce variability and improve manufacturing yield, the SRAM cell now have much straighter and regular lines and corners are much sharper.

OPC reduces variations (1)

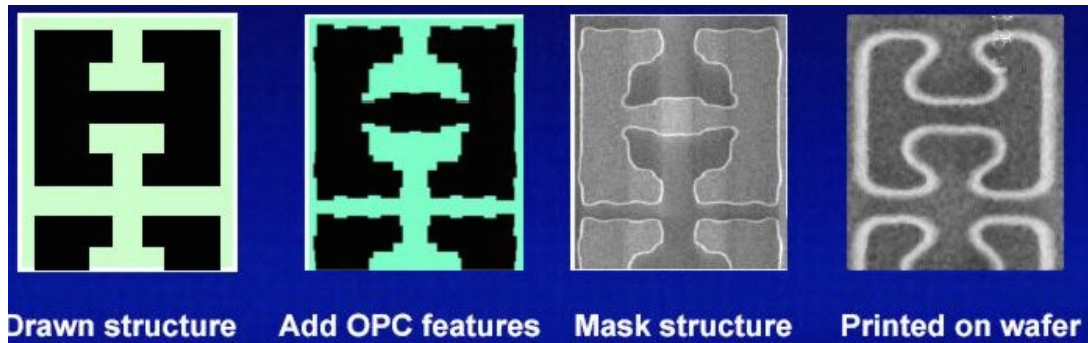


- ◆ **Optical Proximity Correction (OPC)** is a technique to deliberately alter the mask with “roughness” in order to compensate for the non-ideal effect during the fabrication process.
- ◆ The **blue** shape is the desired feature, the **green** shape is the feature AFTER OPC is applied, the **red** shape is the fabricated feature.
- ◆ Corners are deliberately “bulged” in order to reduce the rounding after fabrication.
- ◆ OPC requires high cost in making masks and the electron beam needs to have small spot size and resolution. However, it provides significant improvement in reducing variations due to roughness.

Source: Diagram from LithoGuy, wikipedia

Optical Proximity Correction (OPC) pre-distort the mask images so the the finished features are closer to what are intended. The slide above shows how corners are bulged so that the fabricated feature is more like the read mask.

OPC reduces variations (2)



- ◆ OPC has been crucial from 130nm down to 32nm fabrication
- ◆ Moving to 22nm and below requires other techniques

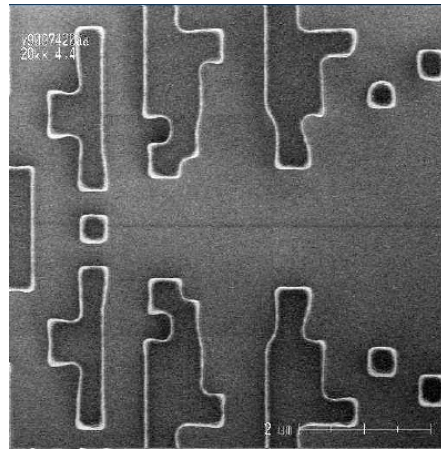
Source: Intel presentation by Singh, ISPD, 2011

This slide shows how OPC can reduce the variations of the mask resulting in final fabricated feature on wafer.

Other lithographical tricks – serif placement



Mask with serifs



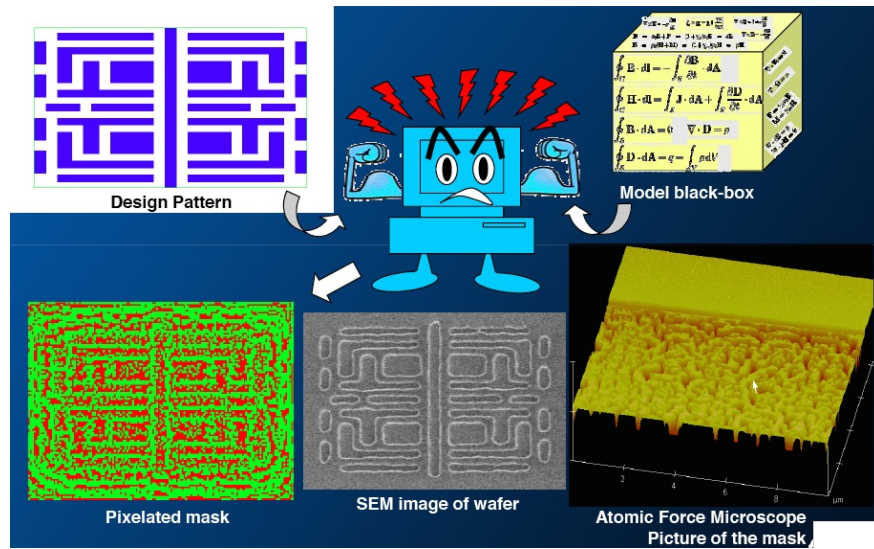
Fabricated

- ◆ Invented by Chen '96 Patent US570765
- ◆ Add little "tails" at corners

Source: Intel presentation by Singh, ISPD, 2011

Another technique proposed was to add serifs or tails to the corners as shown above. This also works.

Other lithographical tricks – Pixelated phase mask

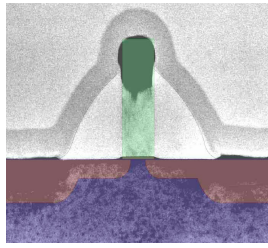


- ◆ Computational lithography – new techniques for further scaling

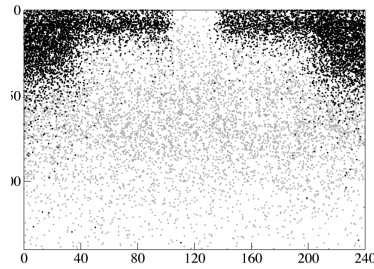
Source: Borodovsky et al., "Pixelated Phase Mask as Novel Lithography RET", Proc SPIE (6924) 2008 and Intel presentation by Singh, ISPD, 2011

Finally, this is another method using pixelated mask originally presented by Intel.

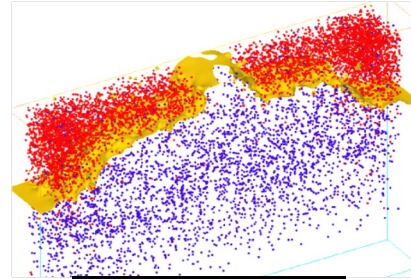
Process variations – random effects in doping



35nm Toshiba trans.



Dopant distribution



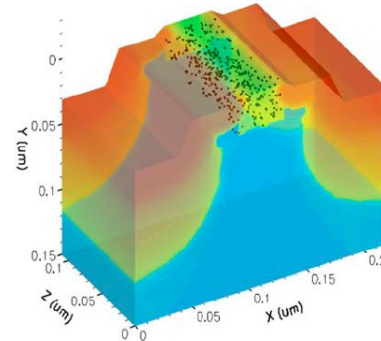
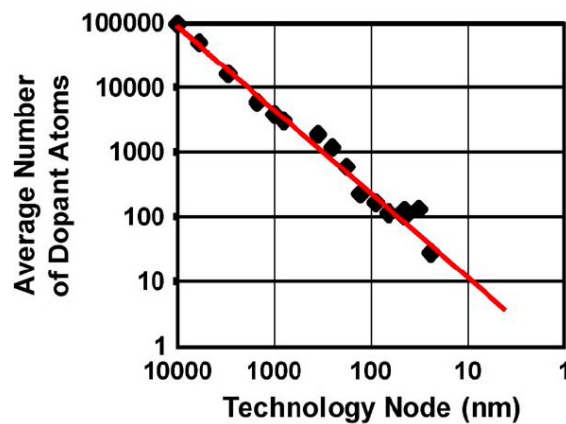
Dopant distribution

- ◆ Transistors are so small – each dopant atom counts
- ◆ Dopant location and density is random
- ◆ This causes random variations in device characteristic

Source: Asenov et al. "Integrated Atomistic Process and Device Simulation of Decanometre MOSFETS", SISPAD'02

As feature size got smaller and smaller, the dopant count drops. The randomness of the dopant distribution starts to play a role in the variation of the

Process variations – random effects in doping

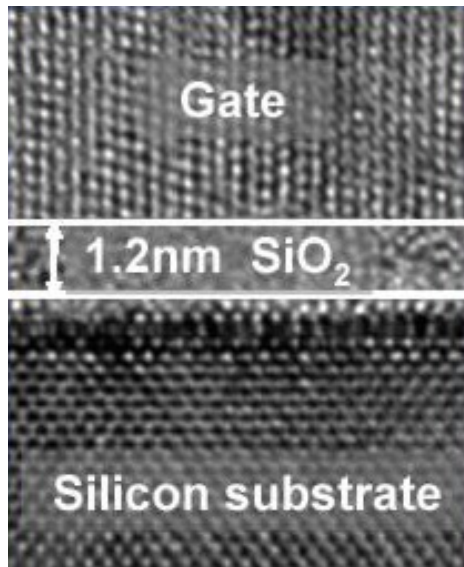


- ◆ Impact of random dopant fluctuation is increasing from generation to generation

Source: Kuhn et al., "Managing Process Variation in Intel's 45nm CMOS Technology" Intel Tech J. (12)2, 2008.

As can be seen in this graph from Intel, the number of dopant atoms may be as low as tens! Missing one or two atom could severely impact on the characteristic of the transistor.

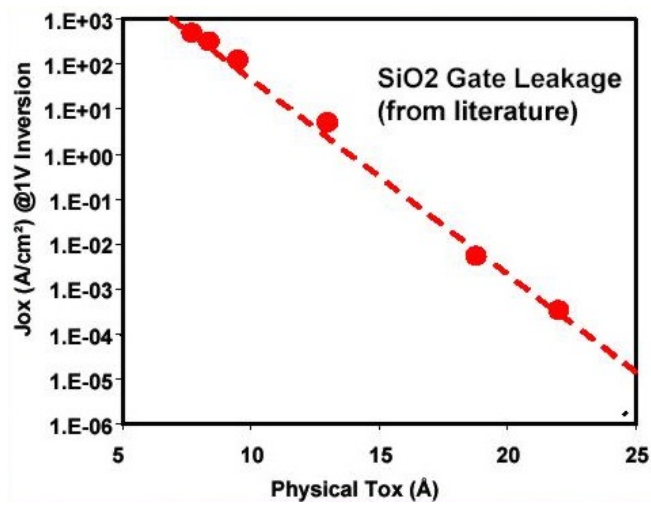
Process variations – very thin gate oxide layer



- ◆ Oxide thickness is now approaching the limit .
- ◆ It is now only around 5 atom thick!
- ◆ This thickness variation is another source of random variations.

Another cause of process variation is due to how thin the gate oxide has become. This slide actually shows the atoms of the gate, the gate oxide and the silicon substrate.

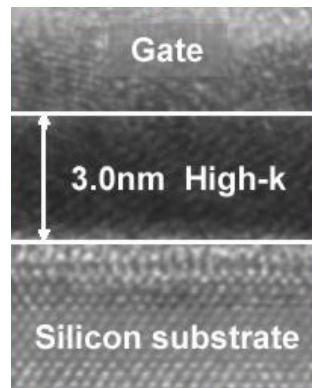
Process variations – thinner oxide leads to larger leakage variations



- ◆ Thinner oxide gives faster transistors
- ◆ Unfortunately also result in higher leakage current
- ◆ Limit of SiO₂ thickness scaling is around 0.8nm

This plot shows how leakage current varies with oxide thickness.

Process variations – High-k dielectric



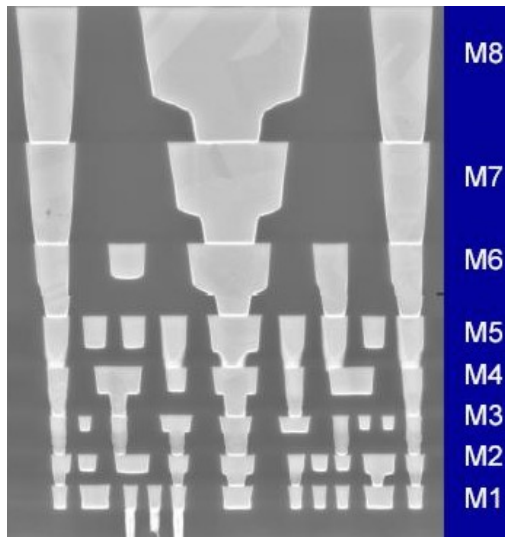
- ◆ Where low leakage is required, use high-k dielectric materials
- ◆ Increase capacitance (hence field strength) by 60%
- ◆ This results in faster transistors
- ◆ Reduces leakage current (thicker), hence >x100 reduction in leakage
- ◆ Yet another new processing step

Source: Intel

High-k dielectric are often used for the gate oxide layer to INCREASE the field strength. The motivation is to reduce the leakage current and make the transistor switch faster.

This is of course not without complication. High-k dielectric requires more processing steps and increase the cost of manufacturing.

Process variations – Interconnect variations

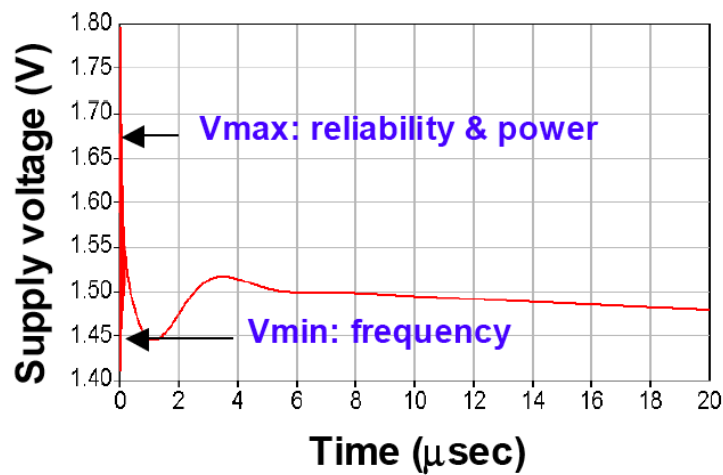


- ◆ 65nm Intel process – 8 layer metal
- ◆ Low-k carbon doped oxide dielectric - reduced capacitance (~ 0.7 of 90nm process)
- ◆ Line and via irregularities also increases random variations

Source: Intel

Unlike gate oxide, interconnect oxide layers use low-k dielectric to reduce interconnect capacitance and crosstalk. This also increase the variations in interconnect.

Supply voltage variations



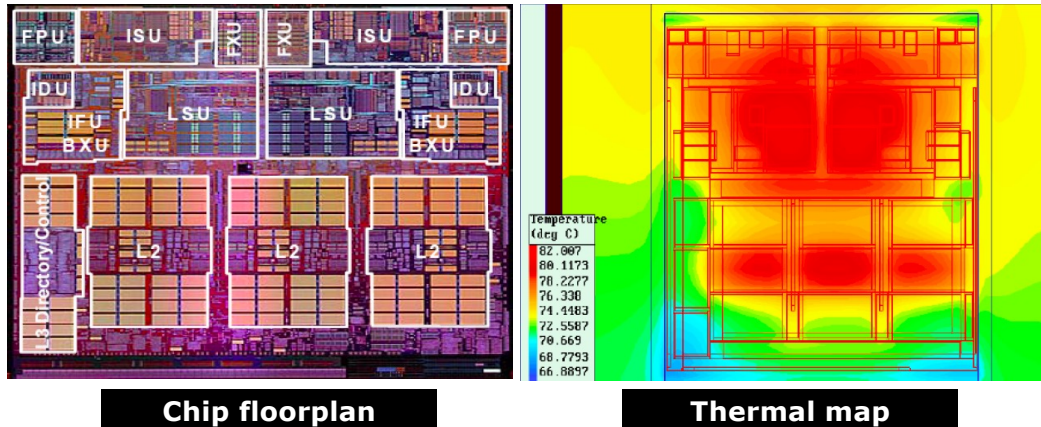
- ◆ Switching activities across the die result in uneven power and IR-drop
- ◆ Also result in local hot spots affecting V_{th} and leakage
- ◆ V_{cc} will continue to scale at around 15% with technology – lower than historical 30%
- ◆ This figure shows how supply voltage drops after chips started warming up

Source: Borkar et al. "Parameter Variations and Impact on Circuits and Microarchitecture", DAC 2003

As we have considered in the previous lecture on interconnection, switching activities result in variation in power supply voltages at different part of the chip. That is the reason why power and ground metal layers are much thicker resulting in lower interconnect resistance.

The graph above shows the power supply voltages changes immediately after power is applied to the chip.

Temperature variations



- ◆ Within-die temperature variations can be large – due to hot spots
- ◆ Here is a PowerPC chip with its temperature map during operation
- ◆ Increased temperature also degrades the hot parts of the chip faster – causing increase temporal variations

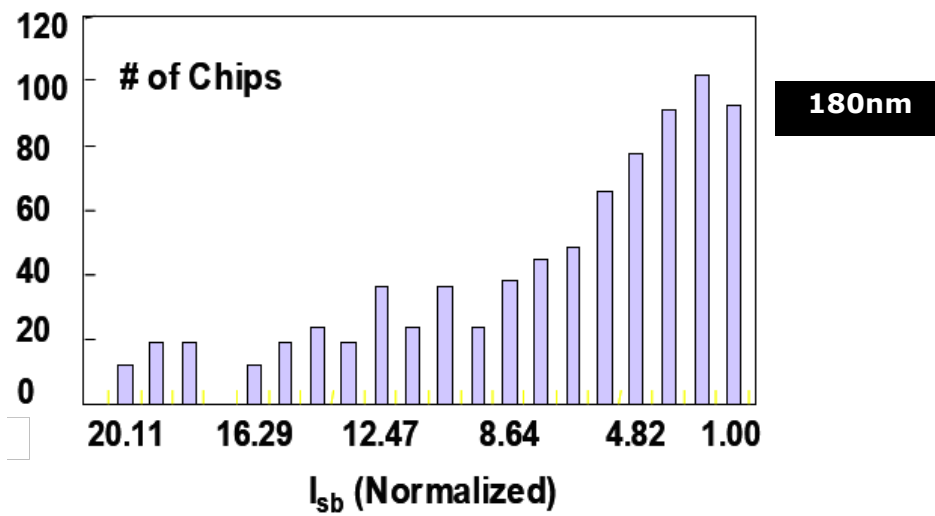
Source: IBM

Different part of the chip experience different switching activities. This results in uneven temperature distribution within the die.

The thermal image of the PowerPC processor shown above demonstrate the local hotspot, resulting in variation of speed and threshold voltages.

Higher operating temperature also causes high degradation or ageing, as will be discuss later in the lecture.

Impact of process variations on I_{sb}



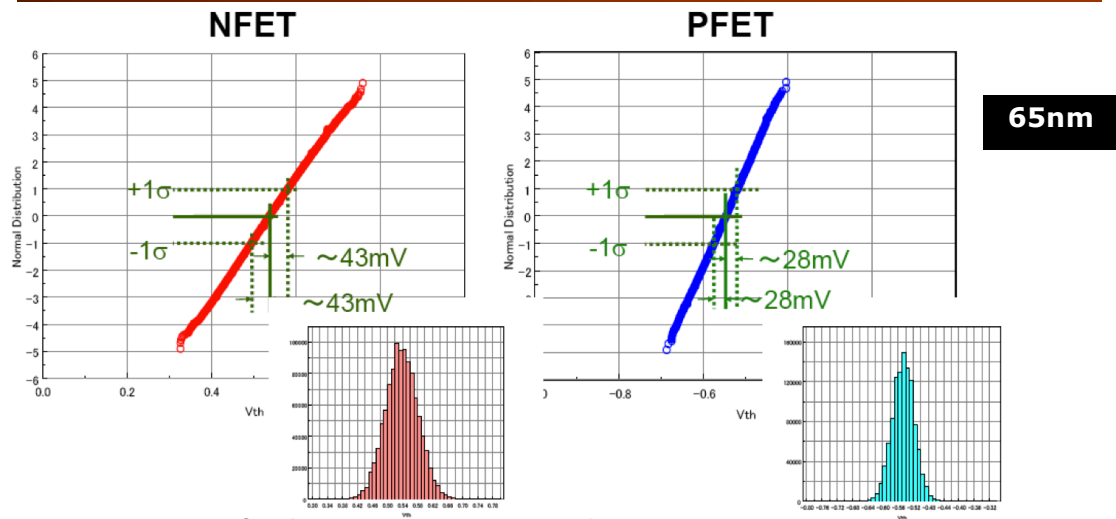
- ◆ I_{sb} variations (substrate leakage current) can be as much as 20x!

Source: Borkar et al. "Parameter Variations and Impact on Circuits and Microarchitecture", DAC 2003

In a CMOS chip, all p-n junctions are reverse biased. The p-substrate is connected to the most negative voltage (i.e. ground).

In spite of the reverse bias junctions to the substrate, there will be leakage diode current to the substrate. This plot shows that the substrate leakage current in a batch of chips. It can vary be as much as a factor of 20!

Impact of variations on V_{th}

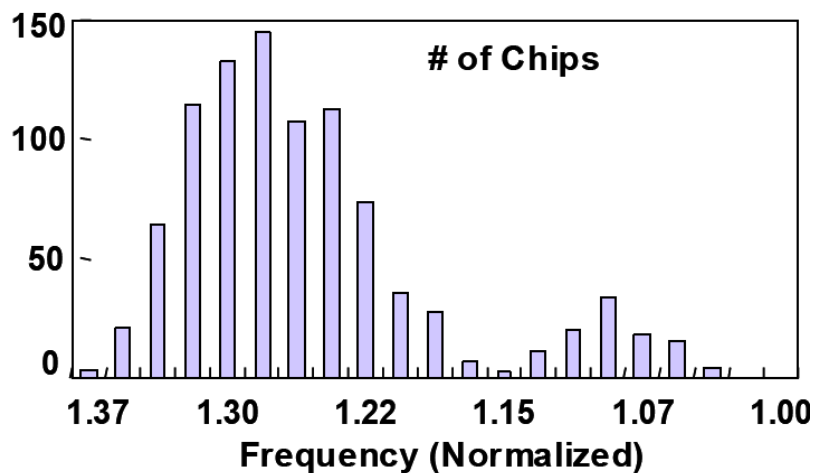


- ◆ Measured V_{th} for 1 million transistors at 65nm
- ◆ Normal distribution up to $\pm 5\sigma$
- ◆ NFET has larger variations than PFET

Source: Tsunomura et al. "Analyses of 5σ V_{th} fluctuation in 65nm MOSFETs", VLSI Symposium 2008

Another significant variation is the threshold voltages of NMOS and PMOS transistors. Tsunomura conducted an experiment in 2008 with 1 million transistors fabricated on a 65nm process – similar to the one we use to fabricate chips for this course. He found that variation of V_{th} could be as much as ± 5 standard deviations.

Impact of variations on operating frequency



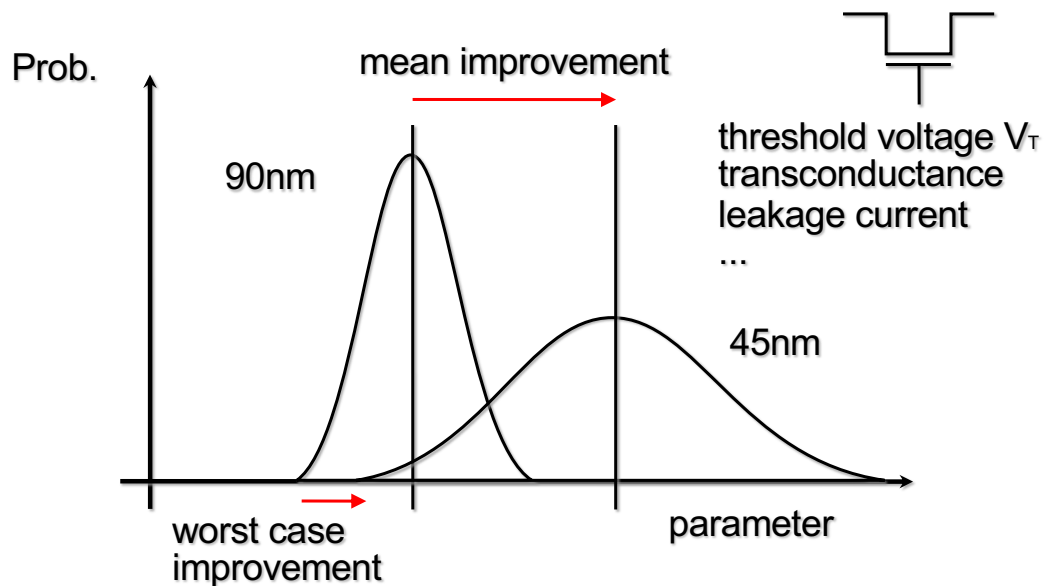
- ◆ As variation increases, the gap between the lowest and highest frequency could be as much as 40% for 90nm.
- ◆ The net effect is reduced the benefit of technology scaling on performance

Source: Borkar et al. "Parameter Variations and Impact on Circuits and Microarchitecture", DAC 2003

This graph shows that all these variations could result in operating frequency varying up to 40% between the highest and lowest frequency. The net effect of this is to reduce the benefit of technology scaling on performance.

Mostly due to increase variations discussed above, the highest operating frequency of processors have been stuck at 3-4 GHz in spite of much improved process technologies with smaller feature size for over a decade.

Variations reduces the benefit of technology scaling

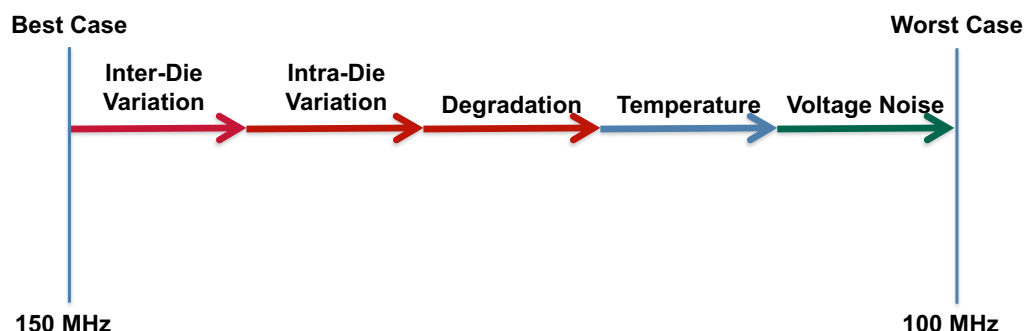


Why are we worry about this sigma value?

Parameter (anything) mean improves as we scale, but spread is getting worse because sigma gets larger. Larger variability.

Since we design to worst case, the actual improvement is much small – could even be worse.

Timing Safety Margin Erosion



- ◆ The consequence of variations is to erode the possible maximum clock frequency from the best case scenario to cope with the worst case
- ◆ The first three factors are process related variations (P)
- ◆ The next two are temperature and voltage variations (T & V)
- ◆ Degradation is also a process variation where the transistors age over time, and gets slower

Let's look in more detail at the reasons why, in conventional design, a large margin is required.

Under the best-case scenario, a circuit may achieve a performance of 150MHz, but the timing model f_{max} is reported as 100MHz.

A certain margin is needed to allow for inter-die and intra die variation.

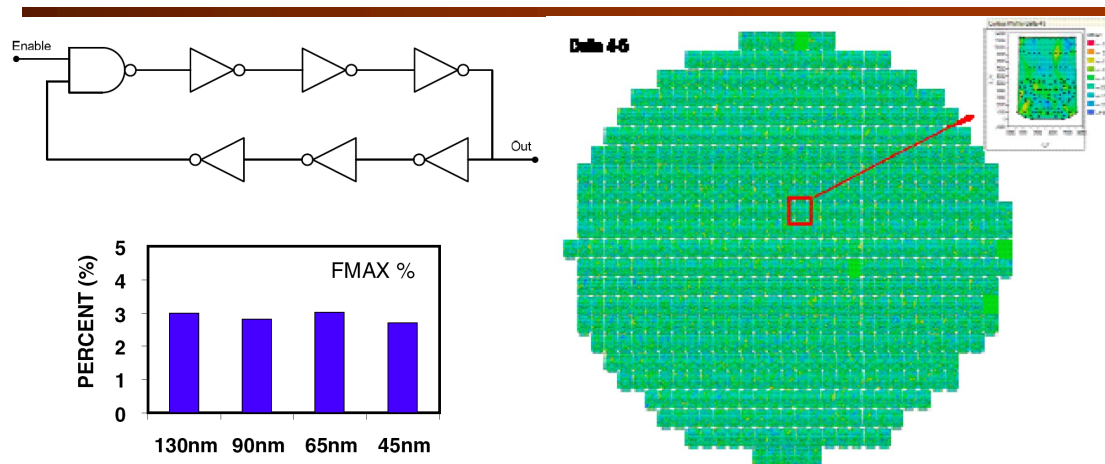
Electronics can get slower with use due to aging, so a further margin must allow for timing degradation.

Another margin is needed to account for temperature variation, both due to ambient conditions, and self heating.

Finally, a margin is needed to account for noise, (cross talk, voltage droop, ripple etc)

The worst case of these factors are accumulated resulting in the model specified worst case of 100MHz

Ring Oscillator – traditional way of measuring variations



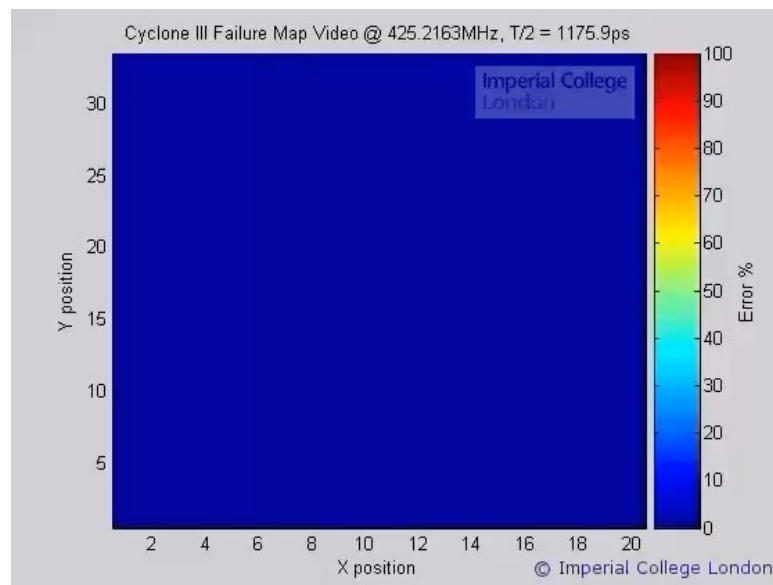
- Using ring oscillators populated with-in die and around the wafer, it is possible to measure variation indirectly through the maximum oscillator frequencies.
- Impact of variations remain around 3% as reported by Intel on their 130nm to 45nm process nodes

Source: Kuhn et al., "Managing Process Variation in Intel's 45nm CMOS Technology" Intel Tech J. (12)2, 2008.

One common method of measuring process variations is to use ring oscillators. Provided that there are odd number of stages of inversion, the circuit shown above WILL oscillate. Assuming each gate delay is the same, and if there are N stages, one can easily derive the unit inverter delay from the oscillation frequency.

The input NAND gate allow the ring oscillator to be enabled or disabled. You are recommended to consider including a ring oscillator with your desing to calibrate the speed of you rchip.

Cyclone III (65nm): Carry chain test movie



Our research team at Imperial College developed an online delay measurement method for FPGAs. This allows us to measure the operating speed of Logic Cells on an FPGA across the chip and study the intra-die speed variation.

This is for the carry chain in cyclone III – 65nm showing those logic cells that fail at different operating frequency on a chip